

Measuring and Mitigating Human Input-Error Perturbations in Language Models

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Abstract

Language models are evaluated on clean text, but humans do not produce clean text: keyboards slip and speech recognizers mishear. We collect a perturbation methodology set that models human input errors in their two dominant modalities, keyboard and voice, and use it both to evaluate models under realistic input noise and to augment training data with the same noise. Applying the set at matched perturbation rates separates the modalities: keyboard errors, which are local and character-level, degrade model performance far less than voice errors, which substitute and merge whole words and so change what the input says. Building on this split, we propose a mitigation that [one-line method description], recovering [XX] of the perturbed-input loss while leaving clean-input performance unchanged. Input-error robustness is a modality property, not a noise-rate property, and it can be trained for.

1 Introduction

A deployed language model never sees the text its user intended. Between intention and input sit two noisy channels: the keyboard, where fingers hit adjacent keys, transpose characters, and drop letters, and the microphone, where a speech recognizer substitutes homophones, merges words, and deletes the punctuation that structure depends on. Benchmarks score models on the intended text; users experience models on the transmitted text. The gap between the two is the subject of this paper.

The two channels corrupt text in structurally different ways. A keyboard error is local: it changes characters inside a word, usually leaving the word recoverable from its neighbors and its shape. A voice error is global: it replaces the word with a different real word, or re-segments a span entirely, so the corrupted input is fluent text that says something else. If models read through local corruption

but are misled by fluent corruption, then robustness to human input error is a property of the input modality, not of the noise rate, and evaluation and mitigation should be organized by modality.

We make three contributions:

1. We collect a **perturbation methodology set** that models human input errors from the keyboard and voice modalities, usable both for evaluation under realistic noise and for training data augmentation (Sec. 3.1).
2. We show that **keyboard errors are less harmful than voice errors** to a language model at matched perturbation rates, and locate why: local character noise leaves meaning recoverable while fluent word-level noise rewrites it (Sec. 3.2).
3. We propose a **mitigation method** that reduces the harm of both input-error modalities, recovering most of the perturbed-input loss without damaging clean-input performance (Sec. 4).

2 Related Work

Robustness to input noise. [cite: character-level adversarial and typo robustness work; synthetic vs natural typo studies; noise injection for NMT and classification. Fetch real citations from arXiv before submission, no invented references.] Prior robustness work concentrates on synthetic character noise or adversarial perturbations chosen to break the model. Human input errors differ on both counts: they are natural, high-frequency, and modality-shaped, and they arrive at inference time whether or not an adversary exists.

Speech-channel errors. [cite: ASR error analysis, homophone confusion, cascaded spoken-language-understanding robustness.] Cascaded systems that feed recognizer output into a language model inherit the recognizer’s confusions, which are fluent by construction: the recognizer emits real words. This places voice errors in a different class

from keyboard slips, a distinction our evaluation makes measurable.

Noise-aware training. [cite: data augmentation with synthetic noise, consistency training, input normalization front-ends.] Augmentation with synthetic noise improves robustness to that noise; the open question our method addresses is which noise to train with when the deployed noise is modality-mixed, and whether robustness to one modality transfers to the other.

3 Findings

We present our findings in two sections. The first builds the perturbation methodology set and establishes that human input errors are structured by modality (Finding 1, §3.1). The second applies the set at matched rates and separates the modalities by the harm they cause (Finding 2, §3.2).

3.1 A perturbation methodology set for human input errors

Finding 1: human input errors are structured by modality. Keyboard errors are local and character-level, leaving words recoverable from shape and context; voice errors are global and word-level, producing fluent text that says something else. The two modalities therefore define two distinct perturbation families, not one noise dial.

Method. We build each family from the physical error process. Keyboard operators model the hands: adjacent-key substitution weighted by key distance on the layout, character transposition, omission, doubling, and shifted-row slips, with per-character error rates calibrated to [typing-corpus source]. Voice operators model the recognizer: homophone and near-homophone substitution, word merge and split, function-word deletion, and punctuation loss, with confusion statistics drawn from [ASR confusion source]. Alongside the twelve modeled operators, two TTS→ASR roundtrip baselines (operators 13 and 14) speak the text through a synthesis engine and re-hear it with a recognizer, giving reference voice corruptions from real engine pairs rather than modeled statistics (Fig. 1; full set in Tab. 1, App. A). Every modeled operator exposes the same severity dial, the fraction of tokens touched, so the two families can be applied at matched rates. The set supports two uses unchanged: perturbing bench-

mark inputs for evaluation, and perturbing training data for augmentation.

The two families do not overlap. Applied to the same text at the same rate, the families produce measurably different corruptions: keyboard operators leave [XX%] of perturbed words out-of-vocabulary and recoverable by edit distance [$\leq k$], while voice operators leave [XX%] of perturbed words as valid vocabulary items, invisible to a spell checker by construction. [One quantitative divergence measure between the families, e.g. distribution over edit types.] A single noise rate therefore under-specifies the perturbation; the modality is the load-bearing variable.

3.2 Keyboard errors harm less than voice errors

Finding 2: keyboard errors are less harmful than voice errors. At matched perturbation rates, keyboard noise costs a language model a fraction of the performance that voice noise costs, across tasks and rates. The harm tracks whether the corruption preserves recoverable meaning, not how many tokens it touches.

Method. We apply both families to the inputs of [benchmark list] at rates [sweep, e.g. 2 to 30 percent of tokens] and score [model list] zero-shot, holding prompts, decoding, and metrics fixed so only the input perturbation varies. Clean-input scores anchor each curve. [n, seeds, SEM protocol.]

Voice noise costs a multiple of keyboard noise. At a matched [X%] rate, keyboard perturbation moves [task metric] from [clean] to [value], while voice perturbation moves it to [value], a [Y×] larger drop; the gap widens with rate. [One sentence on which voice operators drive the harm, e.g. homophone substitution versus punctuation loss, from the per-operator ablation.] The model reads through what a human reader reads through, and is misled where a human would need the audio.

4 Reducing the harm

Method. [The mitigation. Candidate framing to fill in once measured: train-time augmentation with the modality-matched perturbation set at a mixing ratio, optionally paired with an input-normalization step; state the training recipe, budget, and the clean-input control.]

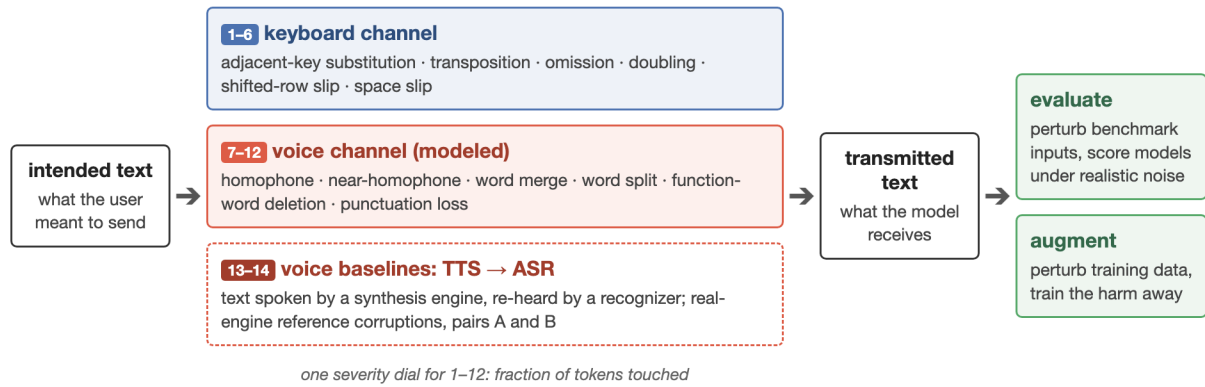


Figure 1: The perturbation pipeline. Intended text passes through one of three channels, the modeled keyboard operators (1–6), the modeled voice operators (7–12), or the TTS→ASR roundtrip baselines (13–14), and the transmitted text feeds evaluation and augmentation. The full operator set is in Tab. 1 (App. A).

The mitigation recovers most of the perturbed-input loss. [Headline numbers: perturbed-input metric before and after mitigation per modality, clean-input metric unchanged within noise, transfer between modalities.] [One sentence on the asymmetry: whether keyboard-trained robustness transfers to voice, and the cheaper direction.]

Limitations. [Bound the claims once results land: models and tasks covered, calibration sources for the error statistics, synthetic-versus-logged human error gap, and any imputed cells, disclosed.]

5 Conclusion

Human input errors are not one noise source but two: the keyboard corrupts characters and leaves meaning recoverable, the voice channel corrupts words and rewrites meaning, and a language model inherits exactly that asymmetry, robust where recovery is possible and fragile where the input is fluent but wrong. The perturbation set makes the asymmetry measurable and trainable: [one-sentence summary of the mitigation result]. Evaluation under clean text overstates robustness for voice-facing deployments, and the fix is to state robustness per modality.

A The perturbation operator set

Tab. 1 lists the full set: the twelve modeled operators with the physical error each models and a worked example, and the two TTS→ASR roundtrip baselines.

#	family	operator	models	example (<i>the road is clear</i>)
1	keyboard	adjacent-key substitution	finger lands on a neighboring key	the roas is clear
2	keyboard	transposition	two keystrokes swap order	the raod is clear
3	keyboard	omission	keystroke missed	the rad is clear
4	keyboard	doubling	key registers twice	the rooad is clear
5	keyboard	shifted-row slip	hand offset by one row	the 4oad is clear
6	keyboard	space slip	space dropped or inserted	ther oad is clear
7	voice	homophone substitution	recognizer picks the homophone	the rode is clear
8	voice	near-homophone substitution	acoustically close word wins	the load is clear
9	voice	word merge	two words heard as one	the roadis clear
10	voice	word split	one word heard as two	the row dis clear
11	voice	function-word deletion	short unstressed word dropped	road is clear
12	voice	punctuation loss	spoken input carries no punctuation	the road is clear no boundary
13	baseline	TTS→ASR roundtrip A	real engine pair [TTS A + ASR A]	[logged output]
14	baseline	TTS→ASR roundtrip B	real engine pair [TTS B + ASR B]	[logged output]

Table 1: The perturbation operator set: operators 1–12 are the modeled methodology set, each with the same severity dial, and operators 13–14 are TTS→ASR roundtrip baselines that produce reference voice corruptions through real engines.